

Hw7

1. $U = ax + by$

$$V = cx + dy$$

$$U = ax + by$$
$$\rightarrow x = \frac{U - by}{a}, \text{ provided } a \neq 0$$

$$V = c \left(\frac{U - by}{a} \right) + dy$$

$$V = \frac{cU}{a} - \frac{cby}{a} + dy$$

$$V = \frac{cU}{a} + y \left(d - \frac{cb}{a} \right)$$

$$y = \frac{V - \frac{cU}{a}}{d - \frac{cb}{a}}$$

$$\rightarrow y = \frac{a(V) - cU}{ad - bc} \checkmark$$

$$x = \frac{U - b \left(\frac{aV - cU}{ad - bc} \right)}{a}$$

$$\rightarrow x = \frac{U(ad - bc) - b(aV - cU)}{a(ad - bc)}$$

$$\ast \rightarrow x = \frac{U(ad - bc) - abV + bcU}{a(ad - bc)}$$

$$\rightarrow x = \frac{U(ad - bc + bc) - abV}{a(ad - bc)}$$

$$\rightarrow x = \frac{U(ad) - abV}{a(ad - bc)} = \frac{adu - abv}{a(ad - bc)}$$

$$x = \frac{du - bv}{ad - bc} \checkmark$$

Condition $\rightarrow$ [The map is invertible
iff the denominator $ad - bc \neq 0$.